

Real-Time DC-Link Capacitor Stress Reduction in Two-Level Three-Phase Inverters Using Adaptive Multi-Carrier PWM Method

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Abstract—This paper presents a novel multi-carrier pulse width modulation (PWM) technique for two-level three-phase inverters to reduce the current stress of the DC-link capacitors, thereby improving system reliability and extending capacitor lifetime. Unlike conventional interleaving approaches used in parallel inverter systems or pulse synchronization in multi-stage converters, the proposed method applies adaptive interleaving between the inverter legs of a single two-level three-phase inverter unit. The carrier phase shifts are dynamically updated in each control cycle based on real-time duty cycles and leg currents, enabling effective redistribution of switching events and minimizing capacitor RMS current. To overcome the implementation complexity associated with high-dimensional input spaces, the method leverages the symmetry and balance of three-phase systems to reduce the number of independent variables. This allows for the creation of a compact look-up table through normalization and transformation, enabling efficient real-time operation. Analytical and simulation results confirm the effectiveness of the proposed PWM strategy in significantly reducing ripple current up to 27% and enhancing the reliability and lifetime of the DC-link capacitors.

Index Terms—Motor drive, DC-link current ripple, sinusoidal pulse width modulation, multi-carrier PWM, voltage source inverter.

I. INTRODUCTION

Two-level three-phase voltage source inverters (VSIs) remain as one of the dominant topologies in industrial motor drives thanks to their reduced component count, simplicity, ease of control, and cost-effectiveness [1], [2]. Despite these advantages, reliability concerns continue to pose a significant design challenge. One of the main sources of reliability concerns is DC-link capacitors, reportedly responsible for approximately 30% of industrial power converter failures [3]. A key contributor to these failures is the ripple current flowing through the capacitor, which induces thermal stress and accelerates degradation, thereby reducing its operational lifetime [4], [5]. Accordingly, various techniques have been investigated to reduce the ripple current magnitude and spectral content to enhance the reliability and extend the lifetime of the DC-link capacitors [6]–[20].

Among the various ripple reduction strategies, interleaving in multi-inverter systems has demonstrated promising results [8]–[12]. Several studies show that optimal phase shifting and current ripple synchronization among parallel inverters

or multiphase machines effectively reduce DC-link ripple and enhance capacitor longevity. Similarly, in two-stage inverter systems, ripple mitigation has been achieved through dynamic coordination of switching sequences between the front-end and inverter stages using interleaved carriers [14], [15].

In contrast to these approaches, which rely on multiple inverter units or cascaded stages, this paper proposes a novel modulation strategy that applies adaptive interleaving between legs of a single two-level three-phase VSI. In the proposed method, leg-specific carrier phase shifts are dynamically updated in each control cycle based on the instantaneous duty ratios and phase currents. This real-time adaptation redistributes the switching instants to minimize the total RMS current stress on the DC-link capacitor, thereby enhancing its lifetime. Notably, this strategy requires no additional hardware, such as parallel inverters or a front-end converter, making it a practical and low-complexity solution for ripple reduction. While the concept builds on interleaving principles, implementing it within a single inverter introduces new challenges due to the time-varying nature of phase currents and the interaction of three duty ratios and three leg currents. Specifically, determining the optimal carrier phase shifts in real time is computationally intensive due to the high dimensionality of the problem. Therefore, the paper introduces a look-up table (LUT), constructed offline, that maps duty ratios and leg currents to optimal carrier phase shifts across all feasible operating points. The comparison of the proposed method with existing literature is summarized in Table I. Unlike prior approaches that rely on parallel inverters, multiphase drives, or additional DC–DC stages, the proposed method achieves effective ripple reduction within a single two-level three-phase VSI. It requires no extra hardware, with the only limitation being the need for an offline look-up table (LUT) to enable adaptive carrier phase shifting in real time.

The remainder of this paper is organized as follows. Section II presents a lifetime estimation model for DC-link capacitors under ripple-induced thermal stress. Section III analyzes the capacitor current behavior in a two-level three-phase voltage source inverter and provides a mathematical formulation for RMS stress evaluation. The conventional single-carrier PWM method is reviewed in Section IV, while the proposed multi-carrier PWM technique is detailed in Section V.

TABLE I: DC-Link Capacitor Ripple Reduction Strategies in Literature

Ref.	Topology	Approach	Evaluation
[11]	Integrated modular drive (IMD)	Multi-phase interleaving	Requires multi-phase IMD; Applying constant carrier phase shifts
[12]	Two parallel 2L VSIs	Hybrid carrier-based PWM	Needs two inverters; Updating reference signals
[14]	Single-phase 2-stage inverter	Carrier shaping in DC-DC stage	Needs pre-stage converter; Altering carrier signals
[15]	Three-phase 2-stage inverter	Generalized carrier-shift method	Needs pre-stage converter; Altering carrier signals
This work	Single 2L-3Ph VSI	Real-time multi-carrier PWM	No extra hardware; Applying adaptive carrier phase shifts

Section VI discusses implementation aspects, including lookup table construction. Section VII presents the comparison of the conventional method and the proposed method, demonstrating the effectiveness of the proposed approach. Section VIII presents simulation results to validate the proposed method. Finally, the conclusion is drawn in Section IX.

II. DC-LINK CAPACITOR LIFETIME ESTIMATION

The lifetime of DC-link capacitors depends on several stress factors, including voltage, current, temperature, and humidity. In this study, a simplified lifetime model commonly used in the literature [21] is employed to estimate capacitor degradation, and is given by

$$L = L_{nom} \times \left(\frac{V_{op}}{V_{nom}} \right)^n \times 2^{\frac{T_{nom}^{hot} - T^{hot}}{10}} \quad (1)$$

where L is the predicted lifetime, L_{nom} is the lifetime under test conditions, V_{op} is the operating voltage, V_{nom} is the nominal voltage, T^{hot} is the operating hotspot temperature, T_{nom}^{hot} is the allowable hotspot temperature under a test condition, and n is the voltage stress coefficient [15]. As shown, the model links thermal stress to lifetime reduction with temperature rise primarily caused by internal power losses. The hotspot temperature, then, can be modeled as

$$T^{hot} = T_{ambient} + R_{eq} \times P_{Loss-ESR} \quad (2)$$

where $T_{ambient}$ is the ambient temperature, R_{eq} is the equivalent thermal resistance, and $P_{Loss-ESR}$ represents power losses due to the capacitors' equivalent series resistance (ESR), which can be calculated as

$$P_{Loss-ESR} = \sum_{n=1}^{\infty} \left(I_{cap,RMS}^2(f_n) \times ESR(f_n) \right) \quad (3)$$

where $I_{cap,RMS}(f_n)$ is the RMS of the DC-link current at the frequency of f_n , and $ESR(f_n)$ is the equivalent series resistance of the capacitors at that frequency. It is also important to note that, as the capacitor degrades, its ESR tends to increase due to wear-out mechanisms in electrolytic capacitors, which further amplifies the effect of ripple current on thermal stress and accelerates the degradation process [22].

III. DC-LINK CURRENT ANALYSIS OF THE INVERTER

Fig. 1 illustrates a conventional two-level three-phase (2L-3Ph) voltage source inverter (VSI) along with its simplified equivalent circuit.

The equivalent model highlights the DC-link current path, under the assumption that all high-frequency components generated by inverter switching are fully absorbed by the DC-link capacitor. Therefore, mathematically, the instantaneous

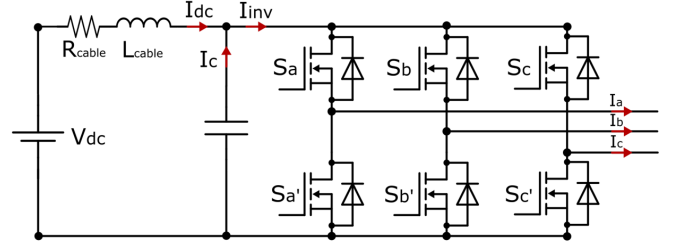


Fig. 1: Two-level three-phase voltage source inverter.

capacitor current ($i_{cap}(t)$) can be expressed as the sum of individual phase currents ($i_x(t)$ for $x \in \{a, b, c\}$) multiplied by their respective switching functions ($S_x(t)$ for $x \in \{a, b, c\}$), minus the average DC component ($\bar{i}_{dc}$) as given by

$$i_{cap}(t) = \sum_{x \in \{a, b, c\}} i_x(t) S_x(t) - \bar{i}_{dc}. \quad (4)$$

The switching function for each phase is determined by comparing the phase reference voltage as

$$v_{x,ref}(t) = m_a \cos(\theta(t) - \frac{2\pi(x-1)}{3}), \quad (5)$$

with the triangular carrier waveform as

$$\text{tri}(2\pi f_{sw}t - \theta_{x,carrier}), \quad (6)$$

and can be calculated as

$$S_x^{conv}(t) = \begin{cases} 1 & \text{if } v_{x,ref}(t) > \text{tri}(2\pi f_{sw}t - \theta_{x,carrier}), \\ 0 & \text{otherwise,} \end{cases} \quad (7)$$

where m_a is the modulation index, f_{sw} is the switching frequency, $\theta_{x,carrier}$ is the carrier phase of legs for $x \in \{a, b, c\}$.

In this paper, a hierarchical approach, covering both switching-period and fundamental-period dynamics of inverter operation, is proposed to analyse the DC-link current. First, at the switching period ($T_{sw} = 1/f_{sw}$), the instantaneous RMS current is computed as:

$$I_{cap,rms}[k] = \sqrt{\frac{1}{T_{sw}} \int_{(k-1)T_{sw}}^{kT_{sw}} i_{cap}^2(t) dt} \quad \text{for } k = 1, 2, \dots, N_{sw}; \quad (8)$$

where $N_{sw} = f_{sw}/f_1$ is the number of switching intervals within one fundamental period. During this period, both the reference voltages and phase currents are assumed to vary

slowly as the switching frequency is much faster than the fundamental frequency, as given:

$$\begin{aligned} v_{x,ref}[k] &\approx m_a \cos\left(\frac{2\pi k}{N_{sw}} - \frac{2\pi(x-1)}{3}\right), \\ i_x[k] &\approx I_{pp} \cos\left(\frac{2\pi k}{N_{sw}} - \frac{2\pi(x-1)}{3} - \theta_{phase-angle}\right), \end{aligned} \quad (9)$$

which allows the capacitor current waveform to be evaluated at quasi-static conditions within each switching cycle. Then, the steady-state fundamental-period RMS current can be calculated by combining all switching-period RMS, as given by:

$$I_{cap,rms}^{total} = \sqrt{\frac{1}{N_{sw}} \sum_{k=1}^{N_{sw}} I_{cap,rms}^2[k]} \quad (10)$$

This formulation shows that minimizing each $I_{cap,rms}[k]$ directly reduces $I_{cap,rms}^{total}$, which means that individual minimization during each switching interval leads to fundamental-RMS minimization.

IV. CONVENTIONAL SINGLE-CARRIER-BASED PWM

The conventional single-carrier-based PWM scheme employs identical triangular carrier signals for all three phases, as presented in Fig. 2. The conventional method's performance is fundamentally constrained by the uncontrolled switching events, which prevent the optimum harmonic cancellation between phases regarding each switching interval. For an example operation, the key waveforms are given as shown in Fig. 3.

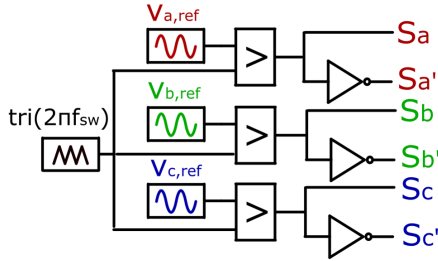


Fig. 2: Conventional Single-Carrier PWM.

V. PROPOSED MULTI-CARRIER PWM TECHNIQUE

The proposed method introduces an adaptive interleaving by applying phase-shifted triangular carrier signals, which are updated within each switching period. The proposed PWM scheme is shown in Fig. 4. Each phase is assigned an individual carrier waveform with a phase displacement δ_x ($x \in \{a,b,c\}$), where $\delta_a = 0$, $\delta_b = \theta_{b,carrier}$, and $\delta_c = \theta_{c,carrier}$. For an example operation, the key waveforms are presented in Fig. 5.

The goal is to dynamically determine the optimal phase shifts $\theta_{b,carrier}^{opt}$ and $\theta_{c,carrier}^{opt}$ for each switching period to minimize the RMS current through the DC-link capacitor. The optimal phase shifts ($\theta_{b,carrier}^{opt}, \theta_{c,carrier}^{opt}$) are obtained

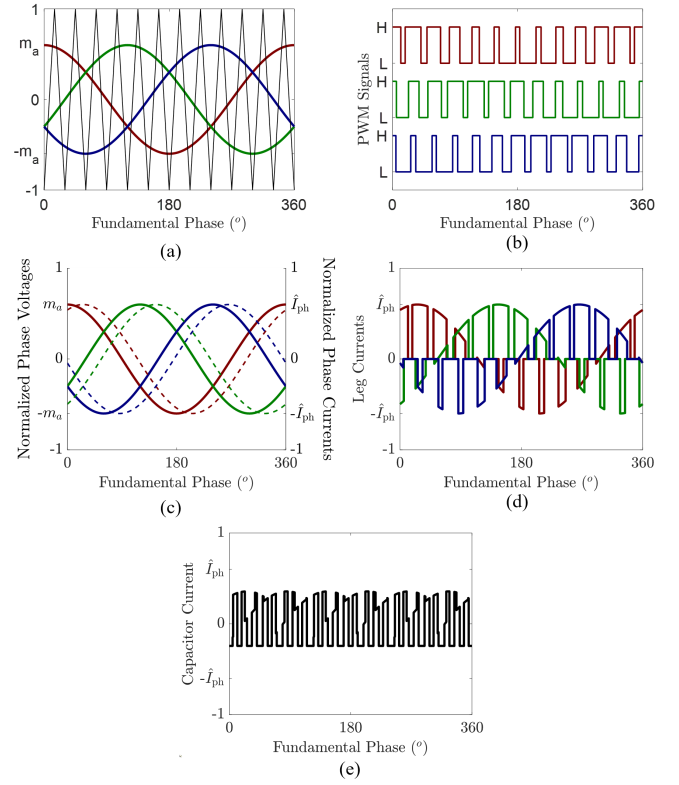


Fig. 3: Key waveforms in conventional SPWM. Red, green, and blue curves correspond to phases A, B, and C, respectively. (a) Reference signals and common triangular carrier waveform. (b) Resulting PWM gate signals for each inverter leg. (c) Normalized phase voltages (solid) and phase currents (dashed). (d) Leg currents. (e) DC-link capacitor current waveform.

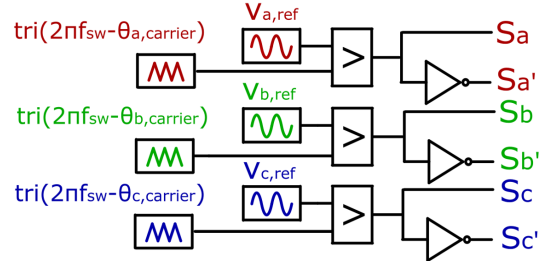


Fig. 4: Proposed Multi-Carrier PWM

by minimizing the capacitor RMS current over the switching period as

$$(\theta_{b,carrier}^{opt}[k], \theta_{c,carrier}^{opt}[k]) = \arg \min_{\theta_b, \theta_c} \left(\int_{(k-1)T_{sw}}^{kT_{sw}} i_{cap}^2(t) dt \right). \quad (11)$$

The optimization is done offline with respect to distinct duty ratios and phase current values. Implementation will be discussed in the following section.

VI. IMPLEMENTATION METHODOLOGY

While the proposed adaptive multi-carrier PWM technique offers significant improvements in reducing DC-link capacitor

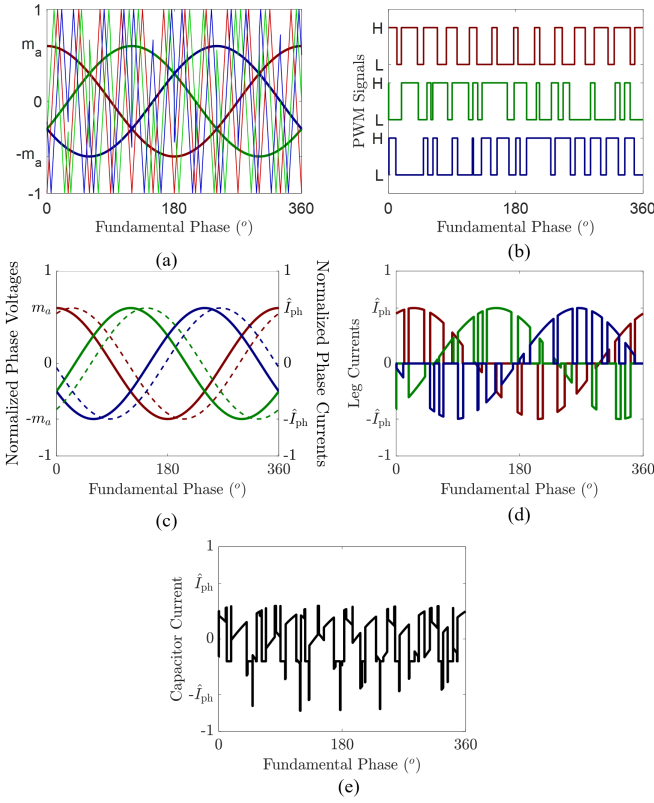


Fig. 5: Key waveforms in the proposed multi-SPWM. Red, green, and blue curves correspond to phases A, B, and C, respectively. (a) Reference signals and triangular carrier waveforms. (b) Resulting PWM gate signals for each inverter leg. (c) Normalized phase voltages (solid) and phase currents (dashed). (d) Leg currents. (e) DC-link capacitor current waveform.

current stress, direct real-time optimization of carrier phase shifts in each switching cycle is computationally prohibitive. To overcome this, an offline optimization and reduced-dimension lookup table (LUT)-based implementation strategy is developed to ensure real-time feasibility.

A. Offline Optimization with Reduced-Dimension Inputs

In each switching cycle, the optimal carrier phase shifts $(\theta_{b,carrier}^{opt}, \theta_{c,carrier}^{opt})$ are determined to minimize the capacitor current RMS over that switching interval. However, performing this optimization in real-time is not feasible due to the high dimensionality and computational cost. To address this, the following steps are adopted:

- **Current normalization:** The instantaneous time-domain phase currents $i_a(t), i_b(t), i_c(t)$ are normalized with respect to the sum of their absolute values:

$$\hat{i}_x(t) = \frac{i_x(t)}{|i_a(t)| + |i_b(t)| + |i_c(t)|}, \quad x \in \{a, b, c\} \quad (12)$$

This normalization provides invariance in the magnitudes of the currents and focuses the optimization on the relative distribution and direction of currents.

- **Dimensionality reduction:** Since the three-phase system is inherently balanced (i.e., $i_a + i_b + i_c = 0$), only two normalized current values are sufficient to describe the system. Hence, the input current space is reduced to:

$$\text{Current Inputs} \mapsto \{\hat{i}_a, \hat{i}_b\} \quad (13)$$

- **Duty cycle reduction:** In SPWM, the duty cycles of the three legs satisfy the constraint $d_a + d_b + d_c = \frac{3}{2}$, allowing any two duty cycles to fully define the modulation state. Thus, the input duty ratio space is reduced to:

$$\text{Duty Ratio Inputs} \mapsto \{d_a, d_b\} \quad (14)$$

Using these reduced sets of variables, the offline optimization can be run to compute the optimal phase shifts $(\theta_{b,carrier}^{opt}, \theta_{c,carrier}^{opt})$ over a grid of operating conditions.

B. Lookup Table Construction

A four-dimensional lookup table is constructed where each entry maps the operating point defined by $(\hat{i}_a, \hat{i}_b, d_a, d_b)$ to optimal phase shifts $(\theta_{b,carrier}^{opt}, \theta_{c,carrier}^{opt})$:

$$\text{LUT} : (\hat{i}_a, \hat{i}_b, d_a, d_b) \mapsto (\theta_{b,carrier}^{opt}, \theta_{c,carrier}^{opt}) \quad (15)$$

The input space is uniformly sampled over practical ranges, and linear interpolation can be used to estimate values between stored points, ensuring continuity and smooth transition across operating conditions.

C. Control Execution

During real-time operation, the control algorithm executes the following steps in each PWM cycle:

- 1) Measure instantaneous phase currents $i_a(t), i_b(t), i_c(t)$.
- 2) Normalize the currents to obtain $\hat{i}_a$ and $\hat{i}_b$.
- 3) Compute the duty cycles d_a and d_b from the voltage references.
- 4) Access the LUT using $(\hat{i}_a, \hat{i}_b, d_a, d_b)$ to retrieve $(\theta_{b,carrier}^{opt}, \theta_{c,carrier}^{opt})$.
- 5) Apply the phase shifts to the carrier signals of legs B and C.
- 6) Generate PWM gate signals using standard SPWM with multiple carrier signals.

D. Benefits of the Approach

The proposed implementation strategy effectively bridges the gap between optimality and practicality. By leveraging current normalization and system symmetry, the input space is significantly reduced, allowing compact and efficient LUT construction. This enables real-time execution without sacrificing the performance advantages of adaptive carrier phase shifting, making the technique suitable for industrial inverter applications with limited computation burdens and high reliability requirements.

VII. PERFORMANCE ASSESSMENT OF THE PROPOSED MULTICARRIER PWM METHOD

In this section, the effectiveness of the proposed multicarrier PWM method is evaluated in terms of reducing the RMS current stress on the DC-link capacitor. The assessment is conducted through numerical calculations. Two distinct operating points are selected for both the conventional and the proposed systems, and the corresponding RMS values of the capacitor current at the switching frequency are calculated. The comparison results are presented in Fig. 6.

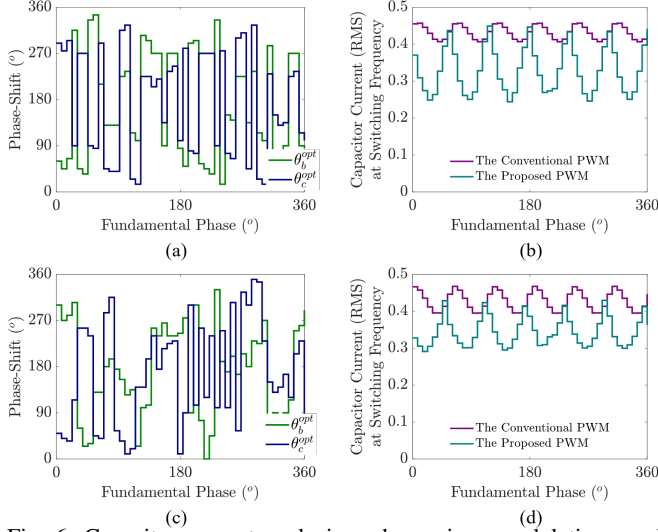


Fig. 6: Capacitor current analysis under various modulation conditions. (a) Optimum carrier phase shift for $m_a = 0.6$, $I_{ph} = 1$ p.u., and power factor $p.f. = 0.9$. (b) RMS capacitor current at the switching frequency for $m_a = 0.6$, $I_{ph} = 1$ p.u., and power factor $p.f. = 0.9$. (c) Optimum carrier phase shift for $m_a = 0.8$, $I_{ph} = 1$ p.u., and power factor $p.f. = 0.95$. (d) RMS capacitor current at the switching frequency comparison for $m_a = 0.8$, $I_{ph} = 1$ p.u., and power factor $p.f. = 0.95$.

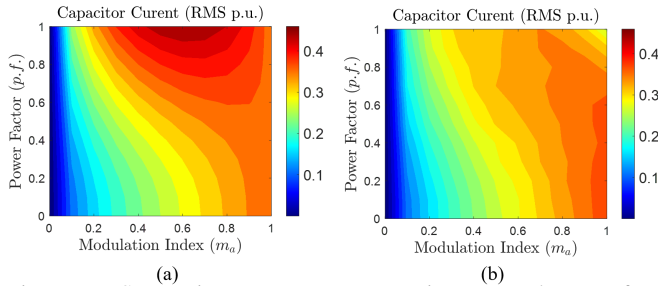


Fig. 7: RMS capacitor current across varying m_a and power factor values. (a) The conventional single-carrier PWM. (b) The proposed multicarrier PWM method

Then, Fig. 7 illustrates the root mean square (RMS) value of the DC-link capacitor current as a function of the modulation index (m_a) and power factor ($p.f.$), comparing the conventional single-carrier PWM with the proposed multicarrier modulation strategy. These results indicate that the proposed method achieves a notable reduction in RMS current (up to 27%) across a wide range of operating conditions. This

reduction leads to decreased thermal stress on the capacitor, which in turn contributes to an extended capacitor lifetime.

VIII. SIMULATION VALIDATION

To verify the effectiveness of the proposed method, time-domain simulations were conducted. A two-level three-phase voltage source inverter with an inductive load was modeled, and the DC-link capacitor current stress was evaluated under both conventional single-carrier PWM and the proposed adaptive multi-carrier PWM strategy.

The simulation parameters are summarized in Table II. The inverter operates at a switching frequency of 50 kHz with a DC-link voltage of 600 V and phase current of $50A_{RMS}$. During simulation, the optimum carrier phase shifts $\theta_{b,carrier}$ and $\theta_{c,carrier}$ were updated at the beginning of each switching period based on the measured phase currents and computed duty ratios.

TABLE II: Simulation Parameters

Parameter	Value
DC-link voltage (V_{dc})	600 V
Switching frequency (f_{sw})	50 kHz
Fundamental frequency (f_1)	100 Hz
Phase current (I_p)	50 A _{RMS}
DC-link capacitance (C_{dc})	50 μ F

A. Comparative Results

Fig. 8 presents the simulation waveforms for both the conventional and the proposed PWM methods with a modulation index of $m_a = 0.6$ and a power factor of $p.f. = 0.9$. Under conventional single-carrier PWM, the DC-link capacitor current exhibits a large RMS value of 27.6 A. In contrast, the proposed multi-carrier PWM method reduces the capacitor current RMS to 21.2 A, corresponding to a reduction of approximately 23%.

Fig. 8 also illustrates the DC-link capacitor voltage ripple for both methods. While the proposed method significantly attenuates the RMS current stress of the capacitor, the adaptive redistribution of switching instants leads to a slight increase in the DC-link capacitor voltage ripple.

B. Dynamic Response Evaluation

To assess the performance under transient conditions, a step change in the modulation index from $m_a = 0.6$ to $m_a = 0.9$ was applied at $t = 0.01$ s, as illustrated in Fig. 9. The results demonstrate that the proposed method maintains lower capacitor current stress throughout the transient period with smooth adaptation of carrier phase shifts with measured current and computed duty ratios.

C. Thermal Stress Comparison

Using the capacitor thermal model described in Section II, the estimated hotspot temperature rise was calculated for both methods where an example point is selected at $m_a = 0.6$ and $p.f. = 0.9$ at rated current. Assuming, as an example, $ESR = 2m\Omega$ and a thermal resistance of 10 $^{\circ}C/W$, approximately estimated using data for the Yageo KEMET C44U series,

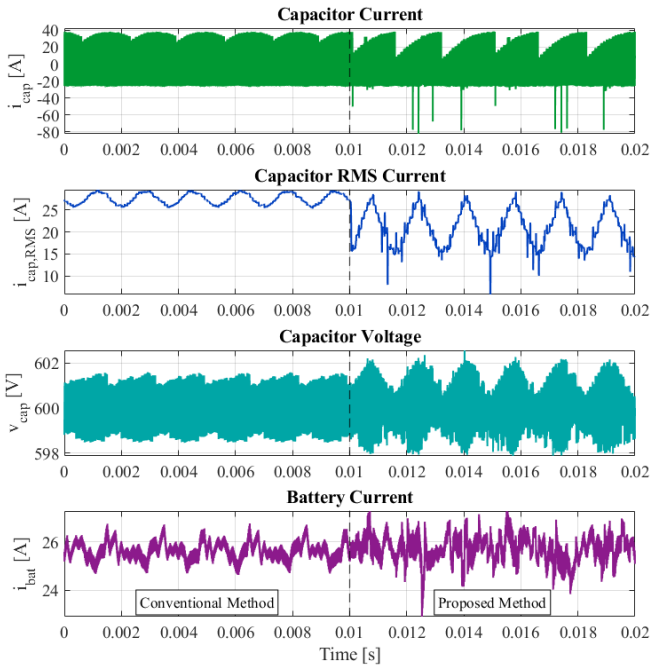


Fig. 8: Simulation comparison of the conventional single-carrier PWM and the proposed multi-carrier PWM methods at $m_a = 0.6$ and $p.f. = 0.9$. From top to bottom: DC-link capacitor current, capacitor RMS current, capacitor voltage, and battery current.

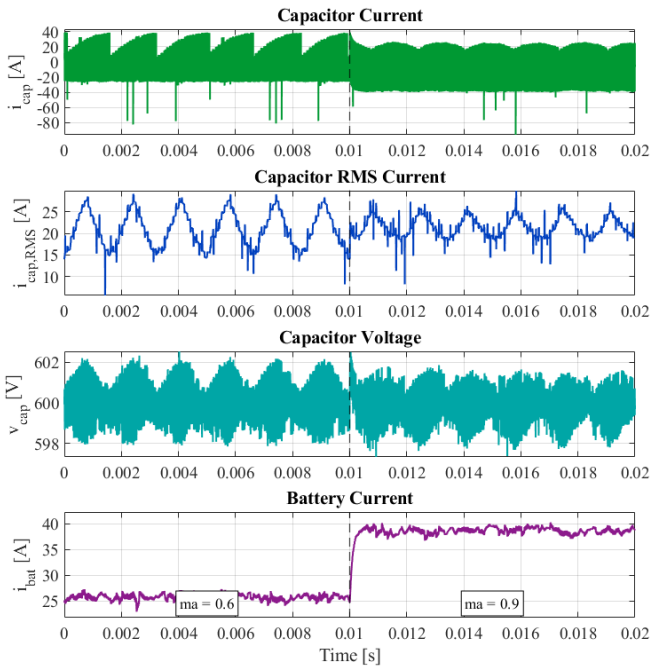


Fig. 9: Simulation waveforms of the proposed multi-carrier PWM method for transition between $m_a = 0.6$ and $m_a = 0.9$. From top to bottom: DC-link capacitor current, capacitor RMS current, capacitor voltage, and battery current.

the conventional method results in a temperature increase of 15.2°C , while the proposed method reduces this to 9°C . Based on the lifetime model, this 6.2°C reduction corresponds to an

estimated lifetime extension of approximately $\approx 50\%$.

IX. CONCLUSION

This paper introduces a novel multi-carrier PWM strategy for two-level three-phase inverters, which aims at reducing the current stress on DC-link capacitors. Unlike conventional interleaving methods that require multiple inverter stages or hardware redundancy, the proposed technique dynamically adjusts the carrier phase shifts of each inverter leg based on real-time duty ratios and leg currents, enabling ripple minimization within a single inverter unit. To address the complexity of real-time optimization, the method utilizes the symmetry of three-phase systems, allowing a reduced input space for a compact and efficient lookup table implementation. Simulation results confirmed a consistent reduction in RMS capacitor current across a wide operating range, with improved thermal performance and extended capacitor lifetime. Overall, the proposed technique offers a low-complexity, hardware-efficient solution for improving reliability and extending the lifetime of DC-link capacitors in industrial inverter applications.

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